

MULTICHANNEL DOOR ALARM WITH DISPLAY

■ PRADEEP G.

A door alarm system alerts you when anyone opens the door of your house to gain entry. It does this by sounding an audible alarm and also displaying the door/gate number which has been opened by the intruder.

Many kinds of alarm systems are available in the market. Here we present a multichannel door alarm with display that can be used to check intrusion through up to nine doors or windows.

The alarm system consists of two main units: door unit and control unit. The door unit consists of a reset switch and a magnetic contact switch (Reed switch). A total of nine door units have been used here. Reed switches are installed on the windows and doors. The alarm sounds when any of these switches disconnects. The control unit controls the alarm and display. The opened door number appears on the display.

Fig. 1 shows the circuit of the door unit. The door unit is made of NAND gates N1 and N2, which form a flip-flop with output pin 4 of N2 connected to input pin 1 of N1 and output pin 3 of N1 connected to input pin 6 of N2. Pin 2 of N1 is pulled low via resistor R9 and pin 5 of N2 is pulled up via resistor R8. Pin 2 of N1 and pin 5 of N2 receive the inputs from switches S2 and S1, respectively.

When the door is closed, reed switch S2 comes near the magnet fitted on the opening shutter of the door. Now press reset switch S1, so that the display shows zero. Reed switch S2 closes and pin 2 goes high.

When anyone opens the door, reed switch S2 opens and pin 2 of N1 goes low. Output pin 3 of N1 goes high to drive transistor T1 into saturation. The

emitter of transistor T1 goes high and LED1 glows. The door unit is extended to the control unit through a three-core cable.

Fig. 2 shows the circuit of the control unit. The control unit comprises

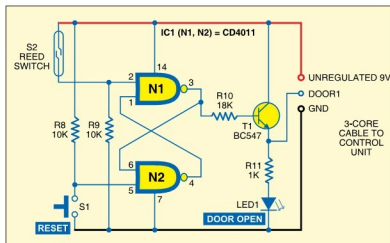


Fig. 1: Circuit of the door unit

PARTS LIST

Semiconductors:	
IC1	- CD4011 quad NAND gate
IC2	- 74LS147 priority encoder
IC3	- 74LS04 hex inverter
IC4	- 74LS247 BCD to 7-segment decoder/driver
IC5	- 7805, 5V regulator
T1	- BC547 npn transistor
D1-D11	- 1N4007 rectifier diode
D12-D20	- 1N4148 switching diode
DIS1	- LTS542 common-anode, seven-segment display
LED1	- 5mm LED
Resistors (all 1/4-watt, ±5% carbon):	
R1-R7	- 270-ohm
R8, R9	- 10-kilo-ohm
R10	- 18-kilo-ohm
R11	- 1-kilo-ohm
RNW1	- 2.2-kilo-ohm resistor network
Capacitors:	
C1	- 1000µF, 25V electrolytic
C2, C3	- 0.1µF ceramic disk
Miscellaneous:	
X1	- 230V AC primary to 9V-0-9V, 500mA secondary transformer
RL1-RL9	- 9V, 1C/O relay
PZ1	- Piezobuzzer
S1	- Push-to-on switch
S2	- Reed switch



encoder 74LS147, seven-segment decoder/driver 74LS247, inverter 74LS04, regulator 7805, relays and a few

discrete components.

IC 74LS147 (IC2) is a priority encoder. It encodes the inputs based on the order in which they are received to ensure that only the highest-order data line is encoded. Its data inputs

and outputs are active at the low logic level. It encodes nine data lines to four-line BCD. When all the nine data lines are at a high logic level, the output of IC2 is '0000.'

All the inputs of IC2 are pulled high via 2.2k resistor network RNW1 and connected to the respective relay contacts. As mentioned before, IC2 converts the input into its BCD complements. The BCD outputs of IC2 ($\bar{A}$ through $\bar{D}$) are fed to inputs A through D of IC 74LS247 (IC4) after inversion by NOT gates N6 down through N3, respectively, and displayed in decimal digit after conversion by IC4.

The 74LS247 has active-low outputs which directly drive the seven-segment LED display. We've used a seven-segment LED display (LTS542) because it is easy to read.

The outputs of the nine door units are connected to relays RL1 through RL9 and diodes D12 through D20, respectively. The cathodes of diodes D12 through D20 are joined together